

AnyCam 3.6.0-rc1.0 Audit Report

Release audit - 04 October 2026 - recording upload by SFTP, FTPS or FTP - tag 3.6.0-rc1.0

Scope: phase 3.6.0 of the build plan: C14, with CrystalHeeler's defaults of 2026-10-04: SFTP, FTPS and FTP; per camera plus global; upload, delete the local copy, retry; passwords encrypted. Built in the untested run on top of 3.5.0-rc1.0. This build is not field-tested.

1. Design

Part	Design
Destinations	A global one (Storage tab, Upload) and, for each camera, global, its own, or off (camera settings, Upload). Kept in /data/upload.json.
When	When a recording stops: <code>_motion_finish_files</code> hands over the files under their final names.
Where	//, the same layout as under /media.
How	Written as .part, then renamed, so a reader never sees half a file. SFTP by <code>asyncssh</code> ; FTP and FTPS by Python's <code>ftplib</code> (FTPS encrypts the data connection too, PROT P).
After	The local copy is deleted unless the destination says to keep it.
Failure	The file stays; the next try is 1 min later, doubling to 30 min. One warning per file, then debug lines. The queue is in /data/upload_queue.json, so it survives a restart.
Test	Uploads a small text file to /anycam_upload_test/.

2. Security review

Risk	Control
Passwords at rest	Encrypted with the same key and code as camera passwords (<code>encrypt_creds</code>).
Passwords to the page	Never sent: the page gets <code>has_password</code> ; an empty password field keeps the saved one.
SFTP server swapped	Trust on first use: the server key's fingerprint is saved at the first upload; a different key stops the uploads and the log says so. Saving the destination again accepts the new key (for a server that was replaced on purpose).
FTP in clear text	Offered because CrystalHeeler chose it; the form warns that FTP sends the password and the recordings unencrypted.
Path tricks	The folder must start with / and must not contain .. ; the server name must have no spaces, slashes, @, ? or #.
New dependency	<code>asyncssh</code> 2.23.1 (with <code>typing_extensions</code> 4.16.0), pinned. 2.24.x needs <code>cryptography</code> 48.0.1; AnyCam pins 48.0.0, so the newest working release was chosen instead of changing the encryption library in an untested build. The release gate now accepts a pure-Python wheel.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates; 17 files; asyncssh and typing_extensions found on PyPI as pure-Python wheels.
Python integration	PASS	560 of 560; 18 new in section AG with stand-ins for ftplib and asyncssh; L13 and Q4 updated for the new route and help text.
JavaScript behaviour	PASS	158 of 158, unchanged; the upload form's 17 element ids all exist in the page.
Undefined names	PASS	17 files, 0.
Not run	NOTED	No field test (CrystalHeeler's order). The image was not built, so asyncssh was not installed or imported for real. No real SFTP, FTPS or FTP server was used. The upload form was not seen in a browser.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*, Parts 1 to 6. Rows this change touches; the rest carry from the 3.5.0-rc1.0 audit.

Standard	Check applied	Result	Note
1.1 Truncated function	AST, contracts, undefined names	PASS	17 files.
1.4 Async	Blocking I/O	PASS	ftplib and file writes run in a thread; SFTP is async.
1.10 Exceptions	Failure paths	PASS	Any upload failure keeps the file and schedules a retry.
1.11 Resources	Connections	PASS	One connection per file, closed in finally or by the context manager.
1.18 Testing	New tests	PASS	18 server checks.
6.1 Security	Input, secrets	PASS	See section 2.
6.2 Dependencies	Pins	PASS	Exact versions; checked online by the gate.
Rule 9 (privacy)	New text	PASS	Tests use example.com-style names only.

Known issues carried forward

- FTP is unencrypted by design.
- A recording uploads only after it stops.
- **B6**: the Microseven waits until CrystalHeeler unlocks it.
- **C21**: the classic engine stays as the fallback.

Acceptance (field test)

- The add-on image builds with asyncssh, and starts.
- Test against an SFTP server: the test file arrives.
- A motion recording arrives on the server under //, and the local copy is gone.
- With the server off, the recording stays and uploads when the server is back.

Files changed

File	Change
anycam_upload.py	New.
anycam_motion.py	Finished files go to the upload queue.
camera_discovery.py	Routes, start-up, version 3.6.0-rc1.0.
page_script.py, anycam_page.py	The upload form; Storage tab and camera settings buttons.
Dockerfile, verify_release.py	asyncssh and typing_extensions; the gate accepts a pure-Python wheel; the new file.
anycam_modules.py, tests/	The new file; section AG; L13, Q4.
CHANGELOG.md, config.yaml, CLAUDE.md, README, docs	3.6.0-rc1.0; build plan.